

SQL Joins

PRIMARY KEY

- A column in a table that **uniquely identifies** each row.
- It cannot contain duplicates or null values.
- e.g. ID in South Africa.

Customer-ID	Name
1	Rofhiwa
2	Lerato

Customers

FOREIGN KEY

- A column that links one table to another table's primary key, creating a relationship between the tables.

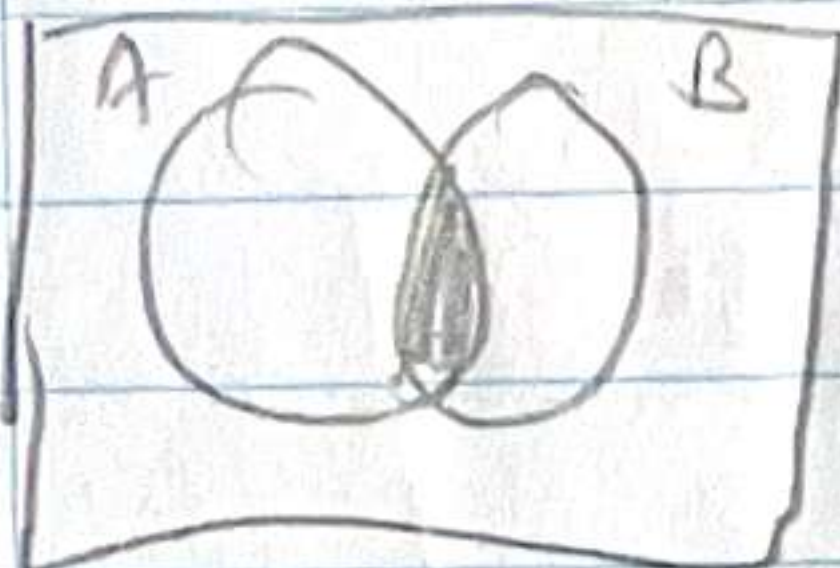
Table: Orders	order-ID	Customer-ID	amount
	101	1	300
	102	1	150
	103	3	500

primary key

foreign key

INNER JOIN

- Allows us to join two or more tables on a common column.
- Gives us the rows that are matching in both tables.



SELECT column 1,
column 2,
column 3

FROM table 1 AS A

INNER JOIN table 2 AS B

ON A.common-column = B.common-column

Customers		Orders		
customer-id	Name	order-id	customer-id	amount
1	Rafhiwa	101	1	300
2	Lerato	102	1	150
		103	3	500

SELECT customer-id,
name,
amount

FROM customers AS A

INNER JOIN orders AS B

ON A.customer-id = B.customer-id;

Customer-id	name	Amount
1	Rafhiwa	300
1	Rafhiwa	150

↳ Only returns Rafhiwa's customer-id because it is the only one common between the 2 tables.

SELECT customer-id,
name,
amount

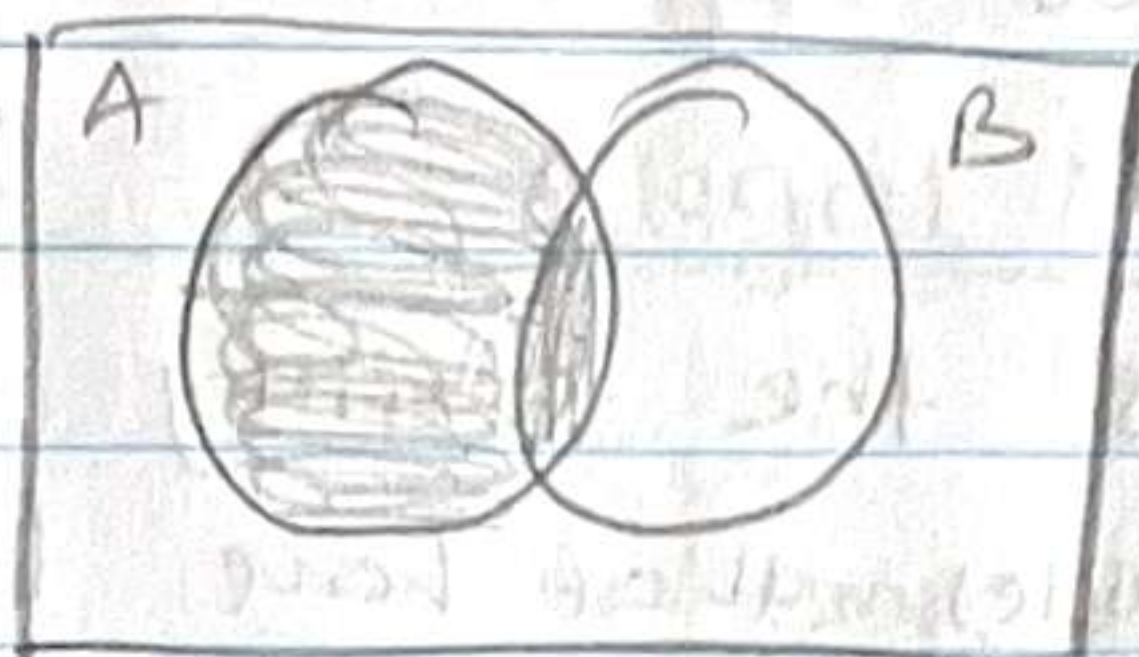
FROM customers

INNER JOIN orders

ON customers.customer-id = orders.customer-id;

Left Join

- ↳ Returns everything from the left / base table and adds matches from the right table.
- ↳ If there is no match, the right table becomes Null.



Select column(s)

FROM table 1 AS A

↳ left table

Left Join table 2 AS B

↳ Right table

On A.common-column = B.common-column;

Select customer-id,
name,
amount

FROM customers AS A

Left Join orders AS B

On A.customer-id = B.customer-id;

customer-id	name	amount
1	Rohiwa	300
1	Rohiwa	150
2	Lerato	Null

* The base table is determined by the question being asked

SQL JOINS

- Used to join two or more tables, needs to have ~~one~~ a common column.
- Joins the affected rows
Columns are not affected.
- Joining tables gives us the ability to select any column between the tables that have been joined together.
- Joined tables become **one**.

student-id	name
1	Ayanda
2	Kabelo
3	Zanele

Students

course-id	Student-id	Course
201	1	Maths
202	3	Science
203	4	English

Course

Left Join

- ↳ Everything from the left table.
- ↳ Only what matches from right-table.
- ↳ If no match, it is null.
- ↳ Outcome will contain all rows in the left table.

SELECT student-id,
name,
course

FROM Students AS A

LEFT JOIN course AS B

ON students.student-id = Course.student-id;

Student-id	name	Course
1	Ayanda	maths
2	Kabelo	Null
3	Zanele	Science
4	Null	English

departments	
department-id	dept-name
10	Sales
20	IT
30	HR
40	Legal

Employees		
emp-id	emp-name	dept-id
1	Alex	10
2	Rafhi	20
3	Thabo	20
4	Lesedi	50

Select dept-id,
 dept-name,
 emp-name, emp-id

From Employees AS A

Left Join departments AS B

On A.dept-id = B.dept-id;

dept-id	dept-name	emp-name	emp-id
10	Sales	Alex	1
20	IT	Rafhi	2
20	IT	Thabo	3
50	Null	Lesedi	4

Students		Clubs		
student-id	name	club-id	student-id	club-name
1	Sam	10	1	Chess
2	Lulu	11	2	Debate
3	Ben	12	4	Soccer

Question: Mention all students that play sports and what sports they play.

SELECT student-id,
name,
club-name

FROM Students AS A

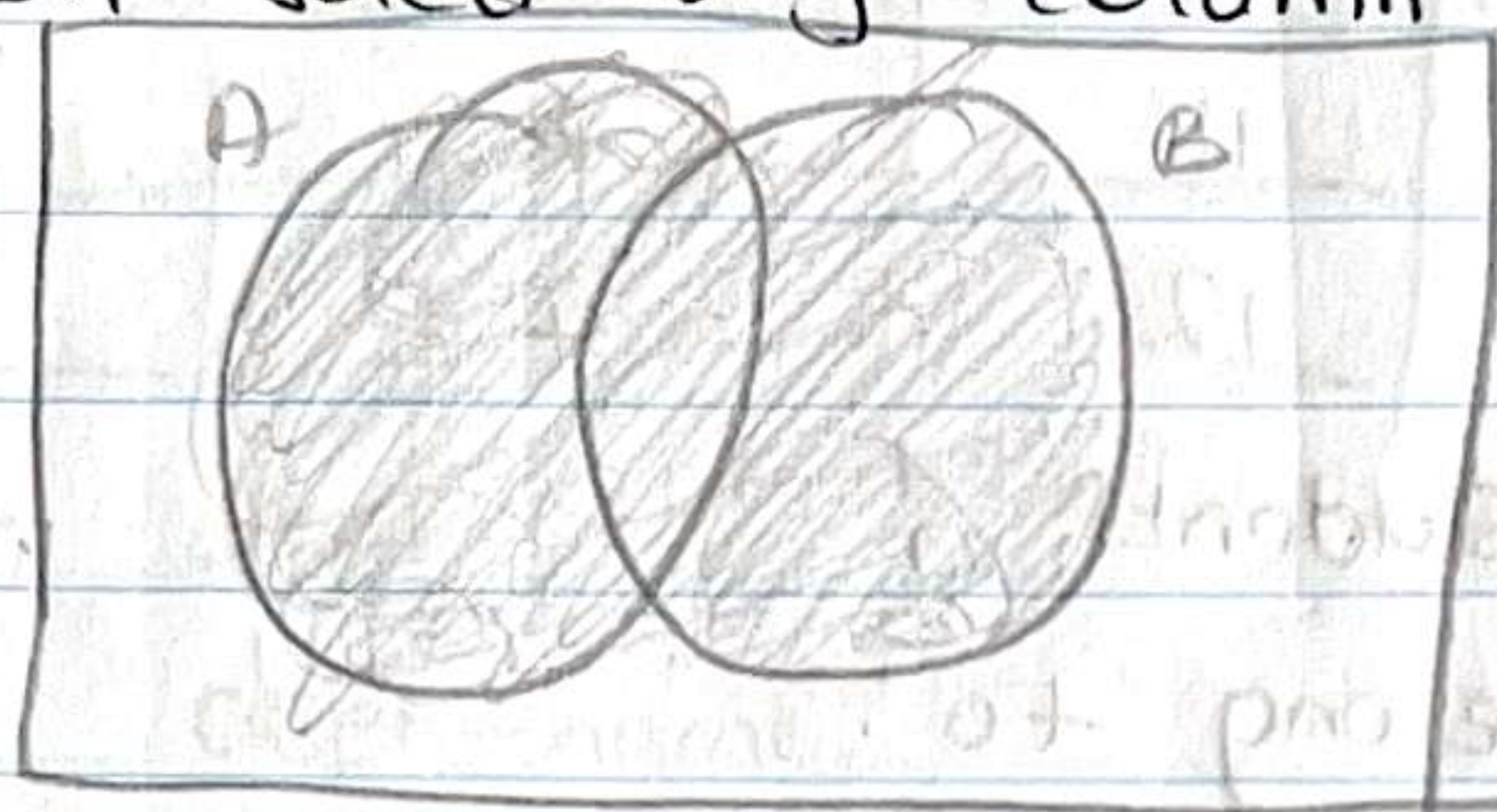
LEFT JOIN clubs AS B

ON A.student-id = B.student-id;

student-id	name Student	club-name
1	Sam	Chess
2	Lulu	Debate
3	Ben	Null

Full Outer Join

- Returns all rows from both tables.
- You can select any column from the tables.



- ↳ Returns everything from both tables
- ↳ If a row matches (has same keys on both sides) then it combines them into one row.
- ↳ If a row exists only in the right table, it will still show up, with nulls for the left table columns.

Syntax:

SELECT * column1
FROM table1 AS A

Full Outer Join table2 AS B

Or A.common-column = B.common-column;

①

Students

student-id	name
1	Sam
2	Lulu
3	Ben

Sports

sport-id	student-id	sports-name
10	1	Chess
11	2	Debate
12	4	Soccer

Get name of students &
club they belong to:

SELECT student-id,
name,
Sport-id,
Sport-name

FROM Students AS A

full OUTER JOIN Sports AS B.

ON A.student-id = B.student-id;

Student-id	name	sport-id	Sports-name
1	Sam	10	Chess
2	Lulu	11	Debate
3	Ben	Null	Null
4	Null	12	Soccer

② Departments

dept-id	dept-name
10	Sales
20	IT
30	HR
40	Legal

Employees

emp-id	emp-name	dept-id
1	Neo	10
2	Aming	20
3	Thabo	20
4	Mia	50

Write a sql script to show all the employees names, and all the department names in the company.

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Select dept-id,  
dept-name,  
emp-id,  
emp-name
```

```
FROM Employees AS A
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```
Full OUTER JOIN Department AS B
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```
On A.dept-id = B.dept-id;
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emp-id	emp-name	dept-id	dept-name
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